

Default Routing:

Default routing is a simplified form of static routing where a router is configured to send all packets destined for networks not explicitly listed in the routing table to a single next-hop IP address or exit interface. This is often used in networks with a single exit point or gateway to the internet.

Key Features of Default Routing

- **Simplified Configuration**- Only one route is needed to handle all unknown destinations, reducing configuration complexity.
- Acts as a catch-all route, directing traffic to a next hop when no specific route matches the destination.
- Commonly used in small networks

Advantages of Default Routing

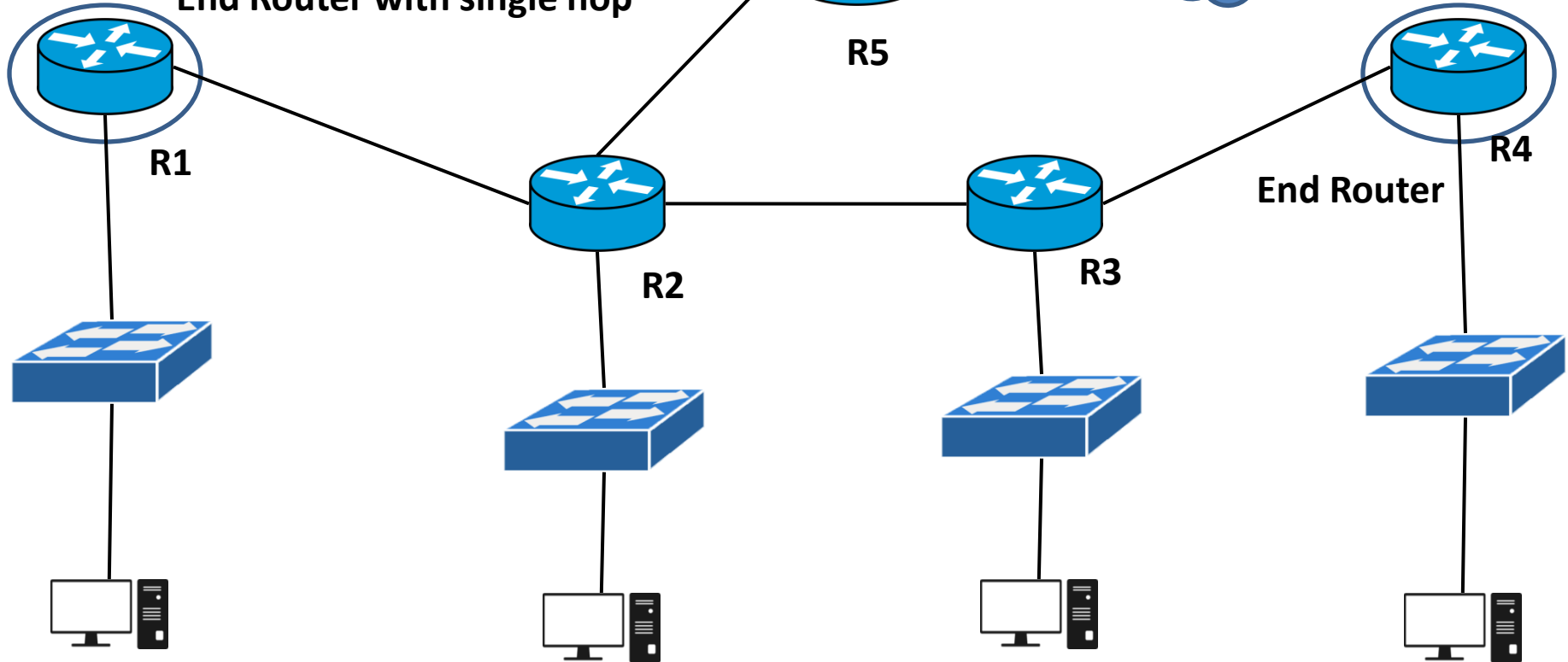
- **Ease of Management:** Simplifies route management by reducing the number of routes that need to be configured and maintained.
- **Resource Efficiency:** Reduces memory and CPU usage on routers compared to maintaining many individual static routes.

Disadvantages of Default Routing

- **Lack of Granularity:** Provides less control over traffic routing, which can lead to suboptimal paths for certain destinations.
- **No Automatic Failover:** Does not provide redundancy; if the default route fails, traffic cannot reach external networks.

Default Routing

End Router with single hop



Default Routing configured on router with single next hop like R1 & R4 (End Routers)

<https://www.youtube.com/@techwithravish>

Default Routing Configuration:

Router> enable

Router# configure terminal

Command for Config:

Router(config)# ip route 0.0.0.0 0.0.0.0 <next_hop_ip>

Example:

Router(config)# ip route 0.0.0.0 0.0.0.0 192.168.1

Router# write memory

Router# show ip route